METRIC
MIL-P-83770B
23 September 1983
SUPERSEDING
MIL-P-83770A
17 December 1974

MILITARY SPECIFICATION

POTASSIUM DICHROMATE - SULFURIC ACID TEST SOLUTION (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of ACS grade potassium dichromate-sulfuric acid solution packaged in 10 milliliter (ml) glass ampoules.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

O-C-265 - Chemicals, Analytical; General Specifications for.
PPP-B-636 - Boxes, Shipping Fiberboard.
PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.
PPP-G-460 - Glass Containers, Filled and Closed; Packaging and Packing of.

STANDARDS

Federal

FED-STD-791 Lubricants, Liquid Fuel and Related Products, Methods of Testing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: San Antonio ALC, Kelly AFB TX 78241, ATTN: SFRM, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6850

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:40Z
Check the source to verify that this is the current version before use.

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extend specified herein. The issues of the documents which are indicated as DOD adopted shall be issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

ASTM Standards

D 941 Test for Density and Specific Gravity of Liquids by Lipkin Bicapillary Pycnometer.

D 1193 Specification for Reagent Water.

D 3951 Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

AMERICAN CHEMICAL SOCIETY (ACS) STANDARDS

ACS Standards

ACS Reagent Chemical Specifications.

(Application for copies should be addressed to the American Chemical Society, 1155 Sixteenth Street, NW, Washington DC 20036.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Traffic Department, 1616 P Street, NW, Washington DC 20036.)

UNIFORM FREIGHT CLASSIFICATION RULES

Uniform Freight Classification Committee

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago IL 60606.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:40Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Requirements. The physical and chemical requirements of the solution shall be as specified in Table I.

TABLE I

REQUIREMENTS

Chemical Requirements	Amount per Liter of Solution	Test
Potassium Dichromate, ACS grade	4.9035 gm ±0.05 gm	4.3.1
Sulfuric Acid, ACS grade	275.0 ml ±15.0 ml	4.3.2
Distilled Water	(the balance)	
Physical Requirements		
Specific Gravity	1.385 ±0.010	4.3.3
Volume	10.0 ml ±0.1 ml	4.3.4

3.2 Ampoule packaging. 10.0 ±0.1 ml of solution shall be packaged in ampoules conforming to requirements of PPP-C-186, Group A, Class 1, Style 7. The ampoule shall be prescored at the neck to provide a location for a clean break.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection Reponsibility. Unless otherwise specified in the contract or purchase order, the supplier has responsibility for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality Conformance Inspection.

4.2.1 Lotting. A lot shall consist of the test solution produced by the manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each homogeneous batch or mix produced from one compounding formulation, in one manufacturing and production operation shall constitute a lot (See 6.3).

4.2.2 Sampling.

4.2.2.1 Inspection. Unless otherwise specified sampling for inspection shall be conducted in accordance with MIL-STD-105.

4.2.2.2 Testing. Sampling for tests shall be conducted in accordance with Table II. Representative ampoules shall be removed from each shipping container and labeled to identify the lot and container from which it was taken.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:40Z
Check the source to verify that this is the current version before use.

TABLE II

SAMPLING

Number of Containers in Batch or Lot	Number of Samples to be taken
2-25	8
26-150	8
151-1200	8
1202-7000	8
7001-20,000	10
Over 20,000	12

4.2.3 Inspection Procedure.

4.2.3.1 For Examination of Preparation for Delivery. The sample unit shall be one filled shipping container, ready for shipment selected just prior to closing operations. Sample containers and the preparation for delivery thereof shall be examined to determine that requirements of section 5 are met. The inspection level shall be S-4 and the AQL shall be 4.0 defects per hundred units.

(a) Contents per container not as specified.

(b) Container not as specified.

(c) Container closure not as specified.

(d) Container damaged or leakage of solution.

(e) Cushioning, partitions, separators, liners, and pads, as applicable, missing or not as specified.

(f) Markings incorrect, missing, or illegible.

4.2.3.2 For Test. Approximately equal portions of all of the specimens taken as directed in 4.2.2 shall be thoroughly mixed together to form a composite specimen of each lot of no less than 80 mililiters. All required tests, as specified in 4.3, will be performed on the composite sample from each lot. Failure of any test shall be cause for rejection of the lot represented.

4.3 Tests. Water in accordance with ASTM D 1193 and ACS reagent grade chemicals shall be used for testing. Blanks shall be run and corrections applied when significant. Tests shall be conducted as follows:

4.3.1 Potassium Dichromate Concentration.

4.3.1.1 Standard Potassium Dichromate Solution. Dry 10 grams(g) of potassium dichromate ($K_2Cr_2O_7$) for one hour at 100°C in a drying oven. Cool 250 ml of water in a 500 ml volumetric flask in an ice bath. To prevent excessive heating, carefully add 187.5 ml of sulfuric acid in small

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:40Z
Check the source to verify that this is the current version before use.

increments, allowing the solution to cool by swirling and placing in an ice bath. Mix well. Weigh accurately 4.90± 0.01g of $K_2Cr_2O_7$ and add to the same flask. Dilute the solution with water to within 10 to 15 ml short of the 500 ml mark. Cook the solution gradually to room temperature, then dilute with water to the mark. Mix the solution well after all the $K_2Cr_2O_7$ has disolved. The solution is stable indefinitely if not contaminated.

4.3.1.2 Ferrous Ammonium Sulfate Solution. Prepare a solution of Ferrous Ammonium Sulfate approximately 0.1N (39.3 g/l + 7 mls H_2SO_4). The normality is not required.

4.3.1.3 Test. Take 10 mls of the standard Potassium Dichromate Sulfuric Acid solution (4.3.1.1) and 10 mls of the solution to be tested and introduce into separate 250 ml Erlenmeyer flasks, add four (4) drops of Ferrion indicator (1.5g of 1, 10-phenanthroline and 0.7g ferrous sulfate dissolved in 100 mls water. To aid in dissolving add 3 mls ACS sulfuric acid). Titrate each to a red end point.

$$\text{gms } K_2Cr_2O_7 = \frac{2.45 \times B}{A}$$

Where A = mls of Ferrous Ammonium Sulfate required to titrate standard
B = mls required to titrate unknown.

4.3.2 Sulfuric Acid Concentration. Take 10 ml of the unknown sample taken in 4.2.3.2 and 10 ml of the standard potassium-dichromate-sulfuric acid solution 4.3.1.1 and place in separate 250 ml Erlenmeyer flasks. To both flasks add phenolphthalein indicator and titrate with 3 N sodium hydroxide solution. Titrate to a red end point (the indicator will override the color of the potassium dichromate).

$$\text{mls } H_2SO_4 = \frac{375 \times B}{A}$$

A = mls of NaOH required to titrate standard
B = mls of NaOH required to titrate unknown

4.3.3 Specific Gravity. The specific gravity shall be determined in accordance with ASTM D 941 at $20^0/20^0$C.

4.3.4 Volume. Each ampoule shall be tested by opening and emptying the contents directly into a graduated cylinder (TC - to contain type) so that the volume measured shall be the same as or more than the labeled amount, 10 mls.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or Industrial (see 6.2).

5.1.1 Level A. Twenty-five ampoules packaged per 3.2 shall be packed in fiberboard boxes confirming to requirements of PPP-G-460. There shall be one tier of ampoules arranged in five rows of five each. A suitable partition equal to the height of the glass containers shall be provided of the halfslotted style. The box shall be lined inside with a suitable

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:40Z
Check the source to verify that this is the current version before use.

fiberboard shell.

5.1.2 Industrial. Ampoules packaged as per 5.1.1 shall be packaged in accordance with ASTM D 3951.

5.2 Packing. Packing shall be Level B or Industrial.

5.2.1 Level B. Twenty-seven boxes of ampoules shall be packed in containers conforming to style RSC, type CF, class weather-resistant of PPP-B-636. Grade W5C shall be used for gross weights not exceeding 40 pounds, and grade V3C for gross weights exceeding 40 pounds.

5.5.2 Industrial. Ampoules packaged as specified in 5.1 shall be packed in accordance with ASTM D 3951.

5.3 Markings. Unit packages, intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended Use. The potassium dichromate-sulfuric acid solution covered by this specification is used as a chemical reagent in determination of the concentration of fuel system icing inhibitor in jet aircraft fuel. (FED-STD-791) and Air Force Petroleum Technical Manuals)

6.1 Ordering Data. Procurements should specify the following:

(a) Title, number, and date of this specification.

(b) Type and size of container required.

(c) Level of packaging and packing required.

(d) Special markings, if required.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to the same physical mixing operation intended to make the final product substantially uniform.

CUSTODIANS:

ARMY-AV
NAVY-AS
AIR FORCE-68

PREPARING ACTIVITY:

AIR FORCE-68

PROJECT NO. 6850-0738

REVIEW ACTIVITIES:

ARMY-MI
NAVY-SH, MC

USER ACTIVITIES:

NAVY-OS

☆U.S. GOVERNMENT PRINTING OFFICE: 1983 — 705-040/4910

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:40Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-83770B	Potassium Dichromate-Sulfuric Acid Test Solution (Metric)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:40Z
Check the source to verify that this is the current version before use.